Fuzzy Tolerance Clustering Workbench

User Guide

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1 About this guide

The workbench is the desktop application of the *fuzzy-tolerance-clustering* project. It runs the possibilistic clustering algorithms of D. A. Viattchenin's D-AFC family — the algorithms described in the book *Fuzzy Tolerance Clustering: Theory, Algorithms, and Implementations* — on your own data, shows the tolerance relation, the allotments and the validity indices, and turns a clustering solution into a fuzzy inference system.

This guide explains the application screen by screen. The mathematics behind the options is only summarised here; the book is the reference for it. The same algorithms are available without a graphical interface as the command-line tool `mfuzzy` and as a .NET library (see Section 13).

Source code, data sets and releases: https://github.com/adamaratski/fuzzy-tolerance-clustering.

2 Installation and first start

Requirements. Windows 10 or 11, 64-bit. Nothing else needs to be installed: the archive contains a single executable with the .NET runtime inside.

Installation.

1. Download `FuzzyToleranceClustering-Workbench-<version>-win-x64.zip` from the *Releases* page of the repository. The file `*.zip.sha256` next to it holds the SHA-256 checksum of the archive.
2. Unpack the archive into any folder. It contains `Math.Fuzzy.Desktop.exe`, the folder `datasets` with the benchmark data sets of the book, `LICENSE.txt` and a short `README.txt`.
3. Start `Math.Fuzzy.Desktop.exe`.

The executable is not code-signed. On the first start Windows SmartScreen may report an unknown publisher; choose *More info* and then *Run anyway*.

First start. The application opens the Dashboard with two sample data sets from the `datasets` folder next to the executable: Anderson's Iris data (150 objects, 4 attributes, 3 classes) and the Yang–Wu data (16 objects, 2 attributes). If the folder is missing, the application starts empty and data can be opened with *Open data*.

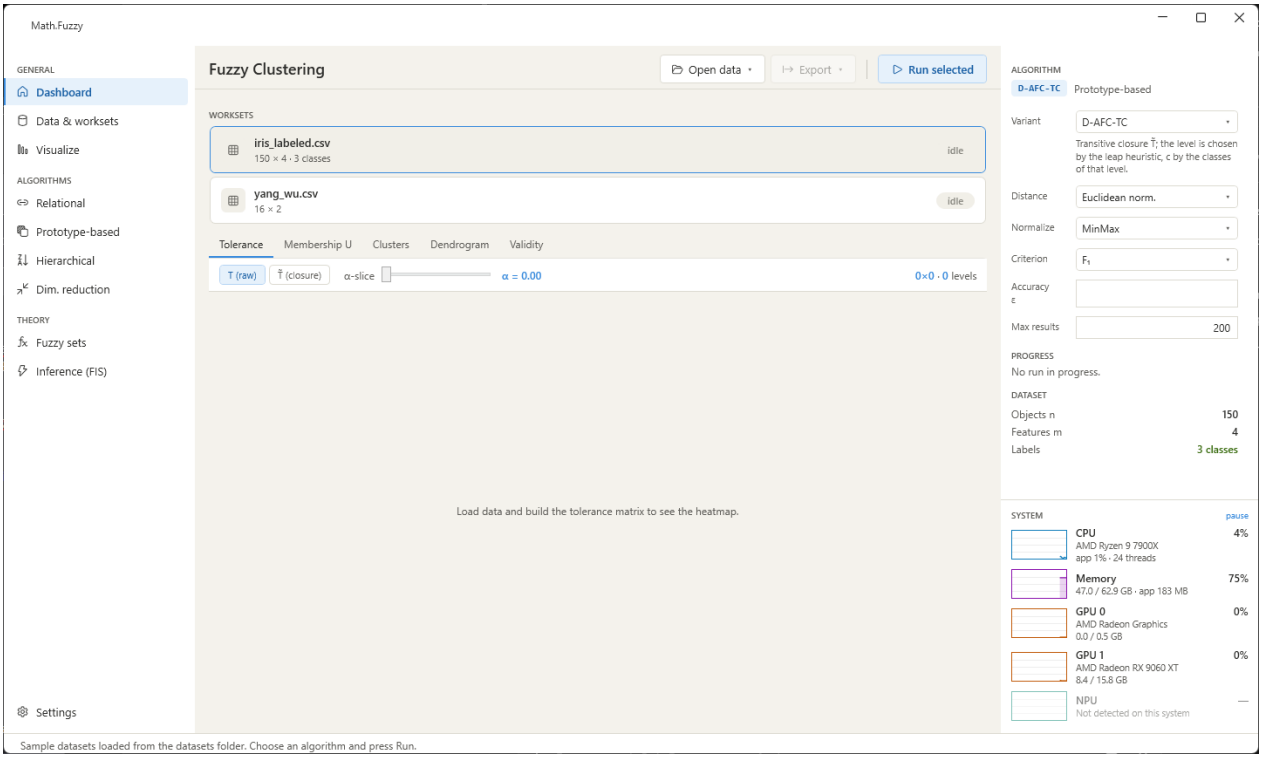


Figure 1: The Dashboard at start-up with the sample data sets.

3 The screen

The window has four areas (see the figure above).

- **Navigation** (left). *General* holds the Dashboard, the data page and the scatter plot. *Algorithms* lists the five categories of algorithms; choosing a category returns to the Dashboard and selects the category's first algorithm. *Theory* holds two pages for exploring fuzzy sets and fuzzy inference.

- **Header** of the Dashboard. *Open data* loads a CSV file or a tolerance matrix, *Export* saves the results of the last run, *Run selected* starts the algorithm configured in the right panel. While a run is in progress the button turns into *Stop*, next to it the algorithm name and the elapsed time are shown, and a progress bar appears under the header.
- **Workspace** (centre). The list of *worksets* — the data sets opened in this session, each with its status (*idle*, *running…*, *done*, *error*) — and the result tabs *Tolerance*, *Membership U*, *Clusters*, *Dendrogram* and *Validity*. Below the tabs a green bar summarises the best allotment of the last run and three tiles show its number of clusters, purity and adjusted Rand index (the last two only when the data set has class labels).
- **Right panel.** The parameters of the selected algorithm (Section 5), the progress of the current run, a summary of the active data set and the *System* monitor with the load of the processor, the memory and the graphics adapters. The monitor can be paused with the small link in its corner.

The status bar at the bottom reports what happened last: a loaded file, the result of a run with its validity indices, or the reason why a run did not start.

4 Loading data

4.1 CSV files

Open data → *CSV data…* (Ctrl+O) opens a delimited text file (comma, semicolon or tab separated; with a semicolon or tab a decimal comma is accepted). The rules are:

- A header row is optional. Column names from the header are used as attribute names on the scatter plot and in the inference page.
- Every numeric column is an attribute. Values use a decimal point.
- One column may hold class labels. A text column is treated as the label column automatically; a numeric column is treated as labels when its header is `class`, `label`, `labels`, `type`, `species`, `variety`, `target`, `category` or `group`. Labels are never used by the algorithms — they serve only to compute purity, the Rand indices and the mutual information after a run. A missing value is reported as an error with its position; a text column whose values are all different (object names) is ignored.

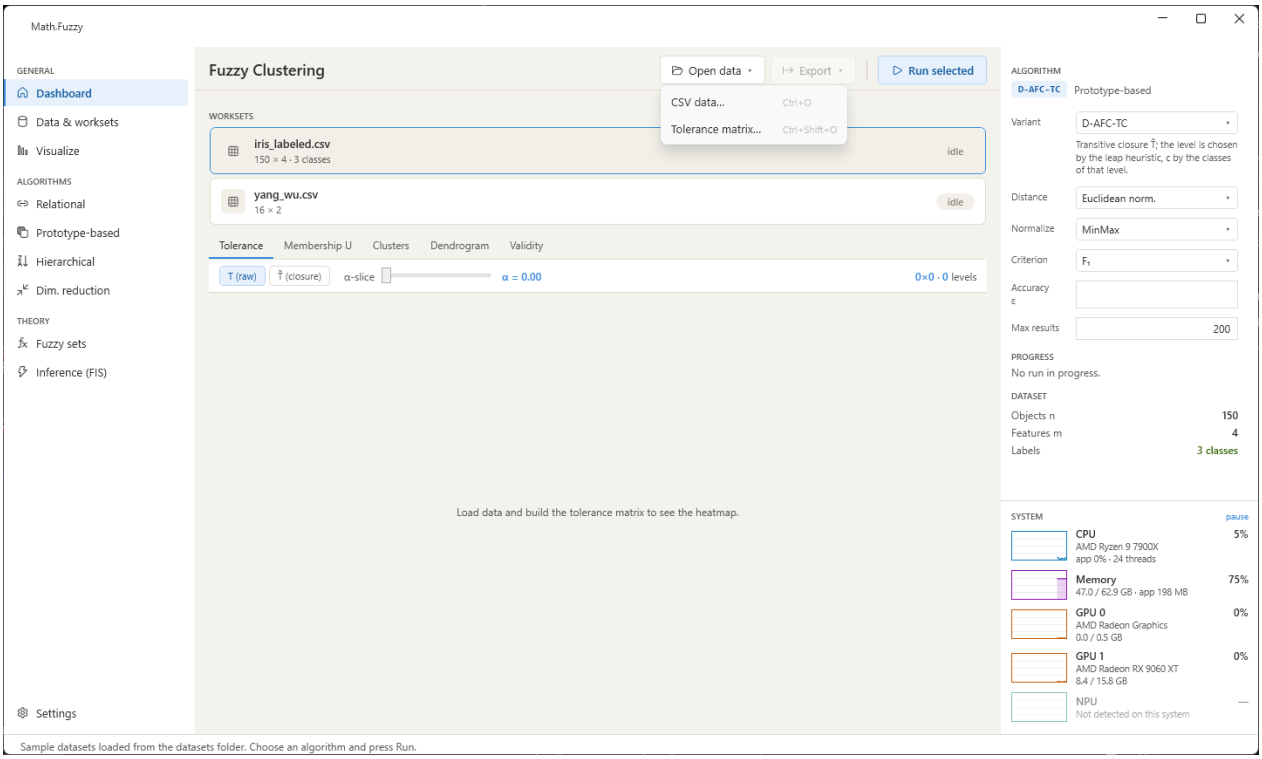


Figure 2: The *Open data* menu.

Every opened file becomes a workset. Click a workset to make it active: the right panel, the result tabs and

the scatter plot switch to it, and a run always applies to the active workset. The data sets shipped with the application have the format described above; the file `datasets/README.md` lists their sources.

4.2 Tolerance matrices

Open data → Tolerance matrix… (Ctrl+Shift+O) loads an $n \times n$ matrix T with entries in $[0, 1]$. The diagonal is set to 1 and the matrix is used as the fuzzy tolerance relation directly, without attributes. The matrix must be symmetric; a header row and a first column of object names are allowed. Every algorithm except the dimensionality reduction runs on such a workset. Since there are no attributes, the prototype-based algorithms then resolve ties between typical points by the object number, and the pages that need attributes (*Visualize*, the inference system) stay empty.

4.3 Intuitionistic fuzzy data

The algorithms of the category *Intuitionistic* (Section 5) work on intuitionistic fuzzy data, in which every degree of membership μ comes with a degree of non-membership ν. There are two ways to open such data:

- *Open data → Intuitionistic tolerance (and matrices)…* loads an intuitionistic fuzzy tolerance. Two file dialogs follow one another: first the $n \times n$ matrix of memberships, then the matrix of non-memberships. Both are read as the tolerance matrix above (a header row and a column of names are allowed). The data set `datasets/hung_ifs_membership.csv` with `datasets/hung_ifs_nonmembership.csv` is an example.
- *Open data → Intuitionistic attribute data…* loads one CSV file whose numeric columns come in pairs (membership, non-membership) per attribute, for example `f1_mu,f1_nu,f2_mu,f2_nu,…`; `datasets/cars10_ifs.csv` is an example. The tolerance is built from the pairs with the similarity measure chosen in the panel (Hamming or Euclidean).

The *Tolerance* tab shows the matrix of memberships of such a workset. If an intuitionistic algorithm is started on ordinary data, or an ordinary algorithm on intuitionistic data, the status bar says what to open or which category to choose.

4.4 The data page

Data & worksets shows the same list of worksets, the size of the active data set and a preview of its first rows. The buttons *Build T* and *Closure T* compute the tolerance matrix and its max-min transitive closure for the active data set with the distance, normalisation and accuracy of the right panel, and display them on the *Tolerance* tab without running a clustering.

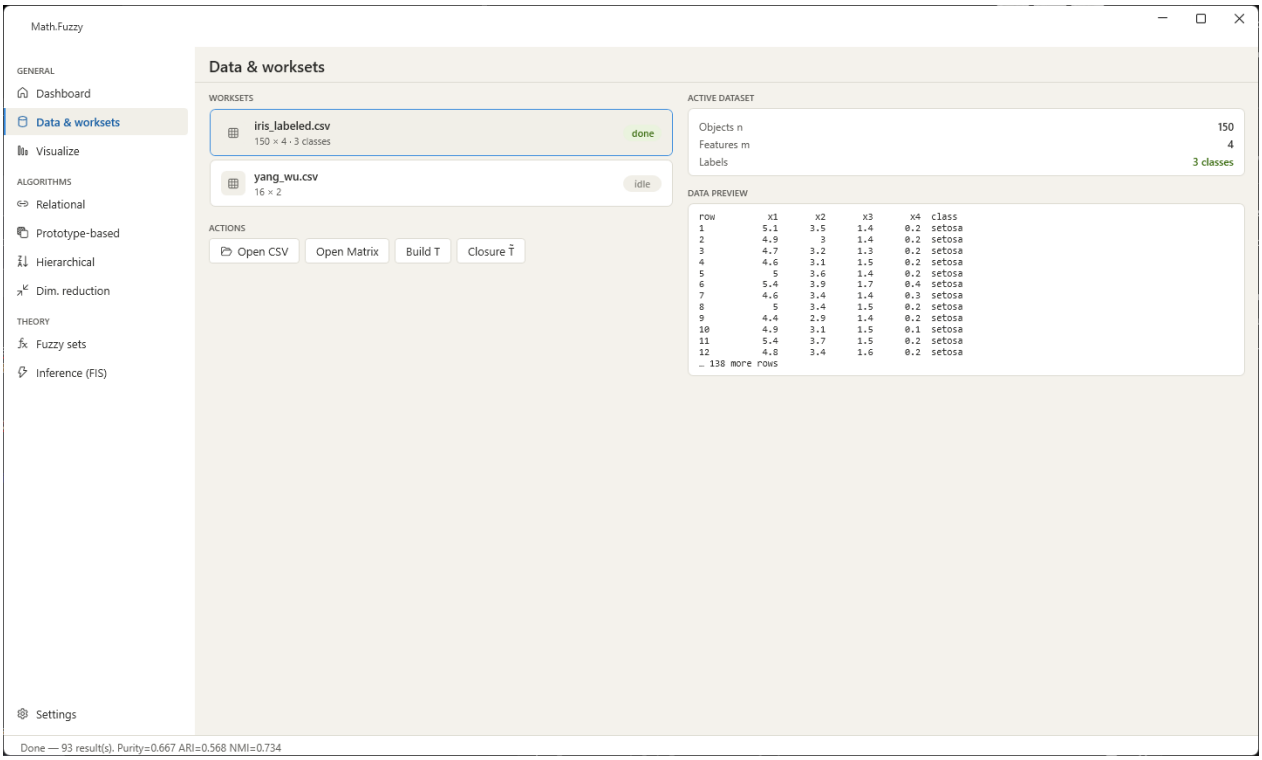


Figure 3: The data page with the preview of the active data set.

5 Choosing an algorithm

The navigation picks a category, the *Variant* box in the right panel picks the algorithm within it, and a short line under the box says what the algorithm does. The blue chip at the top of the panel always shows the selected algorithm.

Table 1: The algorithms of the workbench.

Category	Algorithm	Needs c?	What it does
Relational	D-AFC	yes	The basic algorithm: the best allotment among c clusters over all α-levels of T.
Relational	D-AFC(u)	no	The largest cluster size u is given instead of c; the number of clusters and the threshold follow. The clusters may share objects only if no fully separated allotment exists.
Relational	D-PAFC	no	The principal allotment: the least number of clusters at the first admissible level.
Prototype-based	D-AFC-TC	no	Replaces T by its transitive closure $\widetilde{T}$; the level is chosen by the leap heuristic, the number of clusters follows from the level.

Category	Algorithm	Needs c?	What it does
Prototype-based	D-AFC-TC(α)	no	As D-AFC-TC, but the leap is searched only at levels not below a threshold α^* given by the user.
Prototype-based	D-PAFC-TC	no	The principal allotment on $\widetilde{T}$: its first splitting level.
Prototype-based	D-AFC-TAGA, D-AFC-TAGA(α), D-PAFC-TAGA	no	The same three algorithms with a transitive approximation built by an aggregation operator instead of the closure.
Hierarchical	H-AFC-TC, H-AFC-TAGA	no	The allotments of every level of $\widetilde{T}$ or of the transitive approximation, shown as a dendrogram.
Dim. reduction	D-PAFC-DR, D-PAFC-TC-DR, D-PAFC-TAGA-DR	no	Attribute selection: the same algorithms applied to a tolerance relation between the attributes.
Intuitionistic	D-PAIFC	no	Intuitionistic fuzzy data: the first pair of thresholds (α, β) at which the intuitionistic fuzzy clusters are fully separated, i.e. the least number of clusters.
Intuitionistic	D-AIFC(c)	yes	The number of clusters c is given; the clusters may share objects only if no fully separated allotment exists (the bound w on the pairwise intersections is raised 0, 1, …).
Intuitionistic	D-PAIFC-TC	no	D-PAIFC on the transitive closure of the intuitionistic fuzzy tolerance (max-min on the memberships, min-max on the non-memberships).

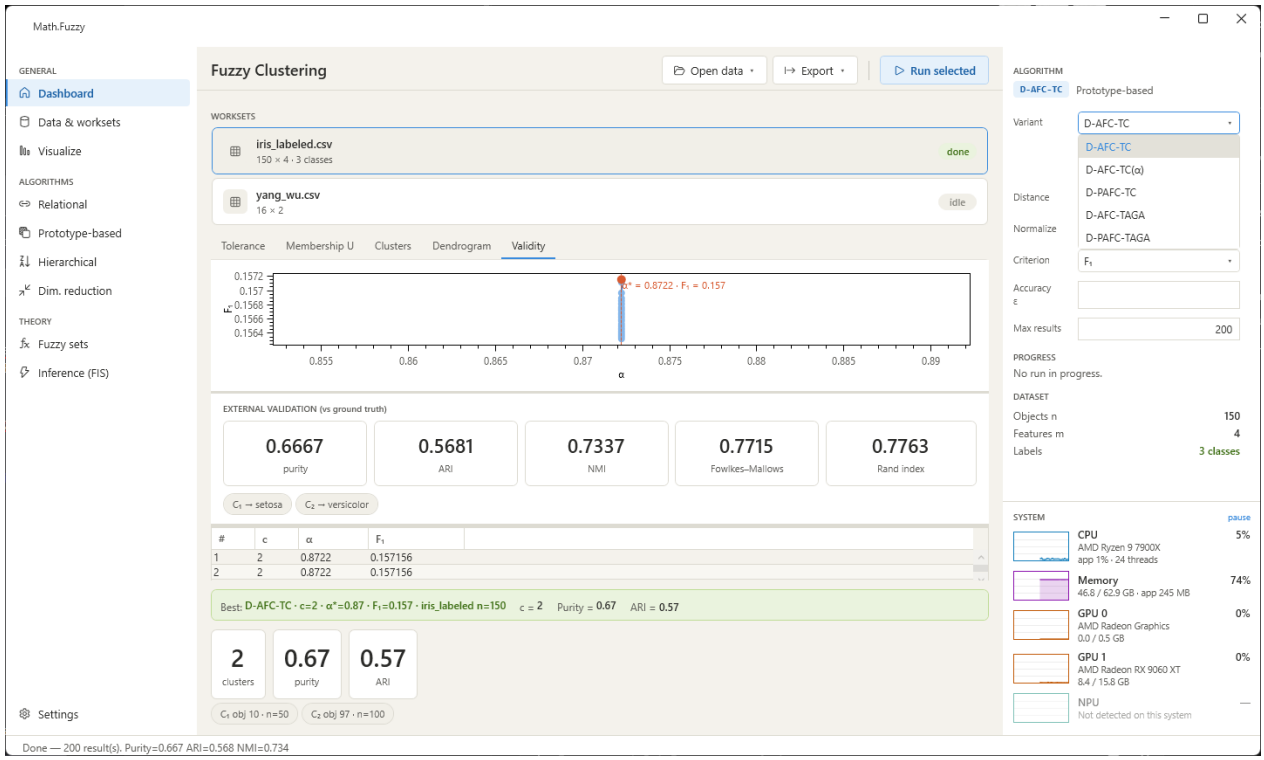


Figure 4: The prototype-based category with its *Variant* box (the screenshot shows version 1.0; version 1.1 adds D-AFC-TAGA( ) to the list).

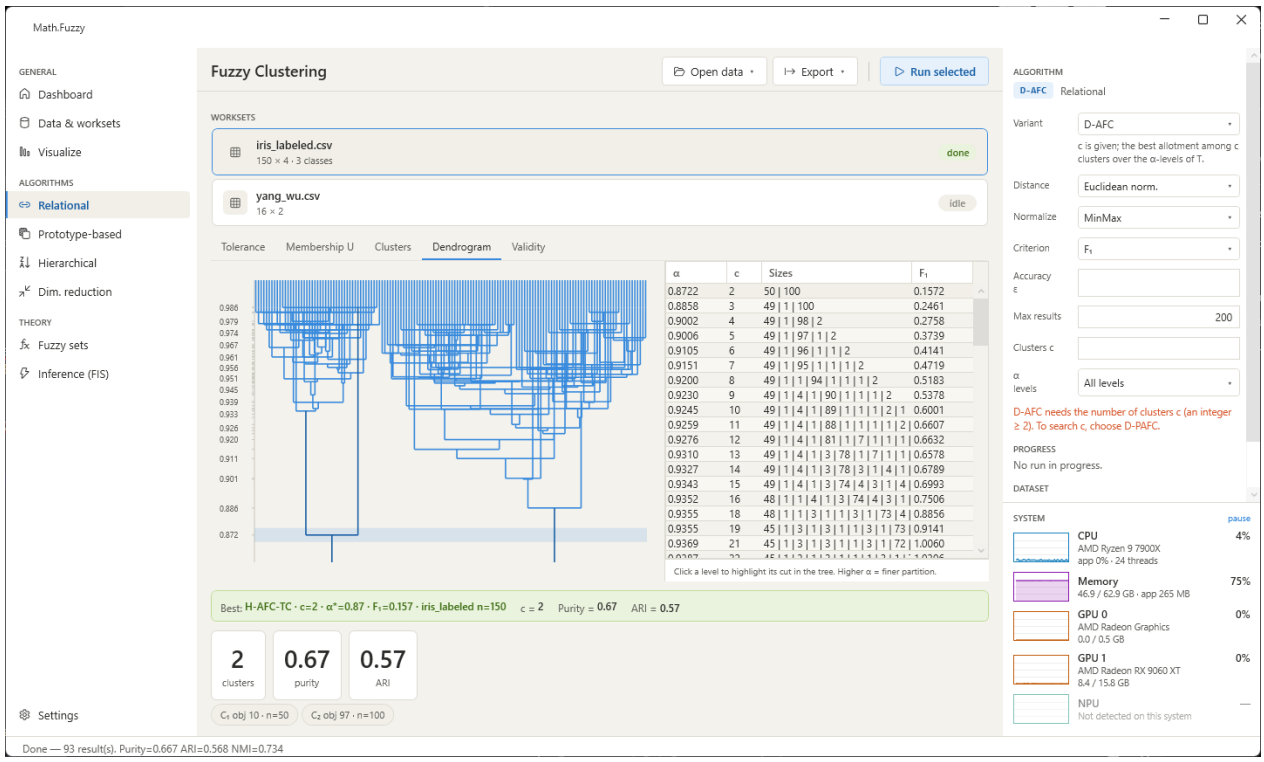


Figure 5: The relational category with D-AFC selected; the fields *Clusters* c and *levels* belong to this algorithm. Until c is entered the panel reports the missing parameter in red.

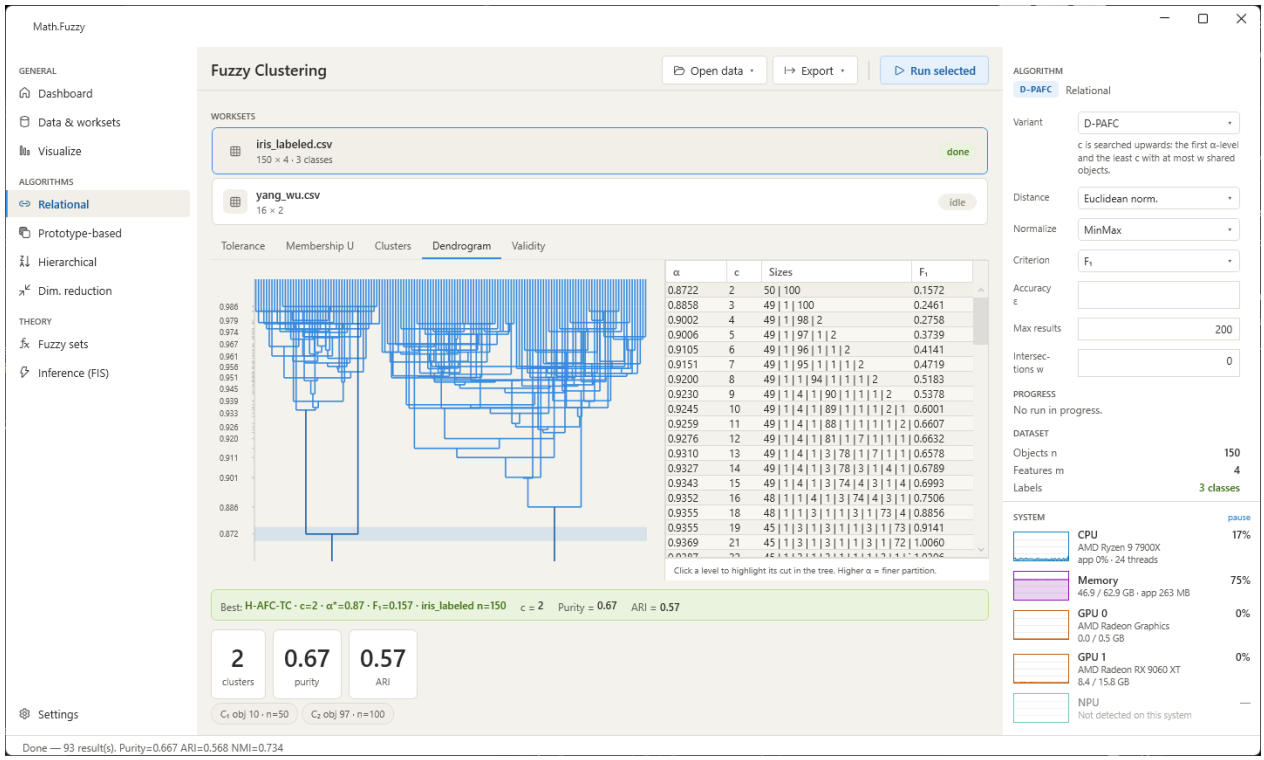


Figure 6: D-PAFC does not need c; instead the field *Intersections* w sets how many objects may belong to more than one cluster.

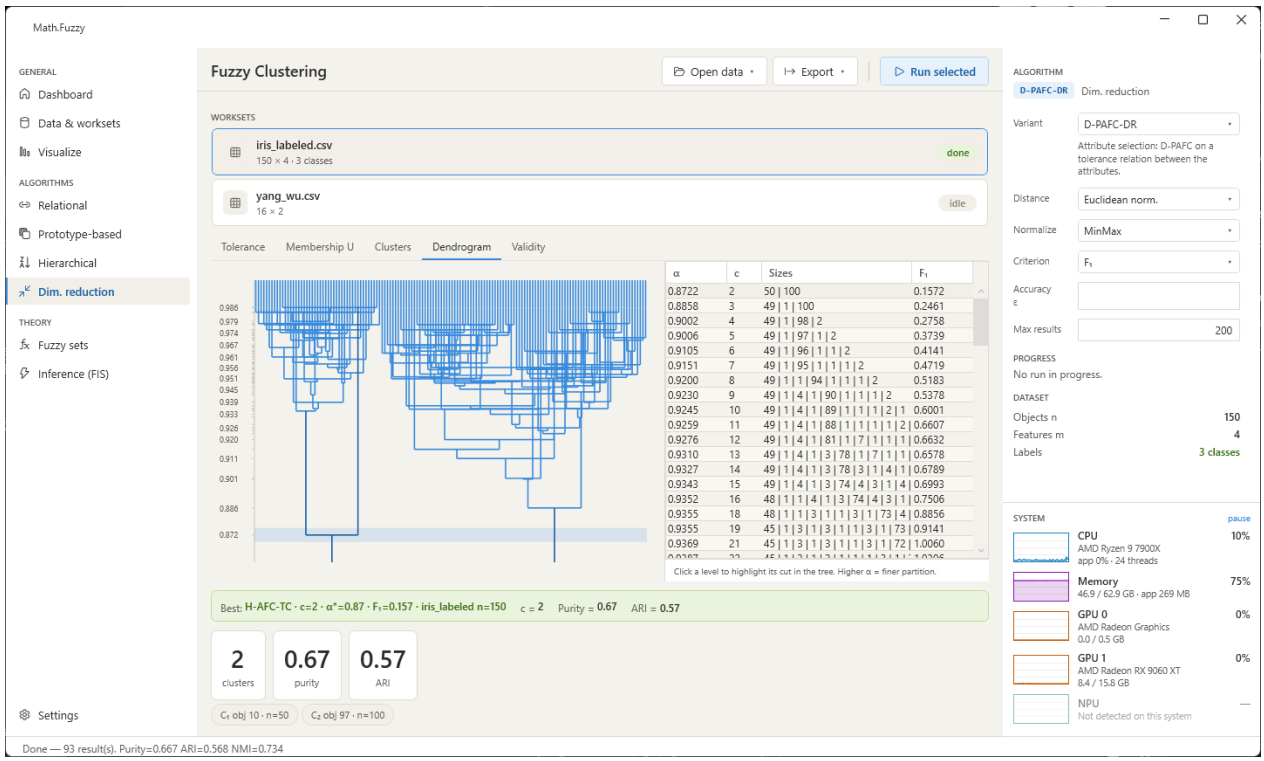


Figure 7: The dimensionality-reduction category.

5.1 Parameters

The upper part of the panel is the same for every algorithm:

- **Distance.** How the dissimilarity of two objects is measured on the normalised attributes: the normalised

Hamming distance, the normalised Euclidean distance, its square, the Manhattan, Chebyshev, Minkowski (with its order p) and cosine distances. The tolerance is $T_{ij} = 1 - d(x_i, x_j)$.

- **Normalize.** *MinMax* maps every attribute onto $[0, 1]$ (unitarisation); *Max* divides every attribute by its largest value (the normalisation of the original experiments); *None* leaves the data as they are.
- **Criterion.** F_1 or F_2, the two criteria by which allotments of the same level are compared; F_1 penalises the number of clusters.
- **Accuracy** . When set, every entry of T is rounded to the nearest multiple of before the levels are built. This merges levels that differ by less than and enlarges the kernels of the clusters. Leave the field empty for full precision. Typical values are 0.00001 to 0.01.
- **Max results.** How many allotments are kept in the result list.

The lower part depends on the algorithm:

- **Clusters c** (D-AFC): the number of clusters, an integer of at least 2.
- **levels** (D-AFC): *All levels* searches every distinct level of T; *Manual* searches a single level: the highest level of T that does not exceed the *** threshold** entered below.
- **Max size u** (D-AFC(u)): the largest number of objects a cluster may have, an integer below the number of objects.
- *** threshold** (D-AFC-TC(α), D-AFC-TAGA(α)): the lowest admissible level.
- **Intersections w** (D-PAFC): the number of objects allowed to lie in more than one cluster; 0 asks for fully separated clusters.
- **TAGA operator** (TAGA variants): the aggregation operator of the transitive approximation — maximum (the transitive closure), minimum, mean, median, upper and lower median. *Automatic* computes all six approximations and takes the one nearest to T.

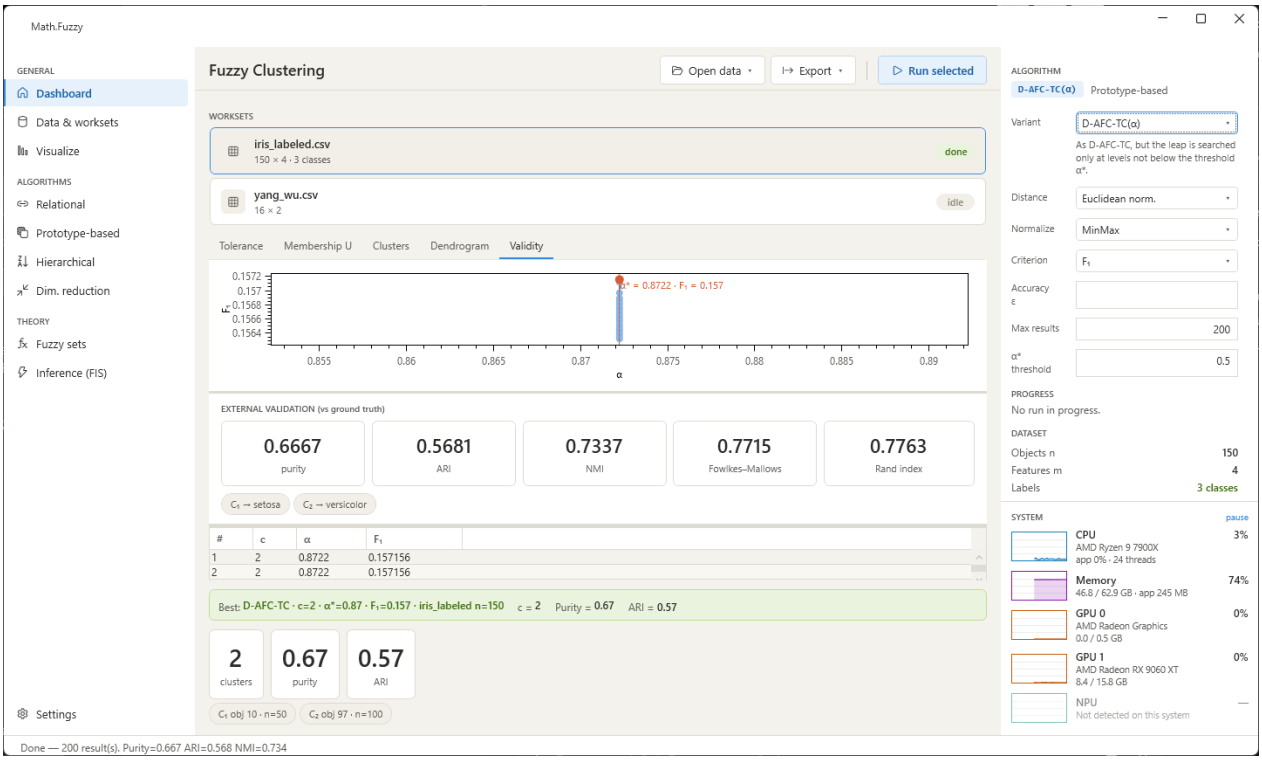


Figure 8: D-AFC-TC( ) with its field ** threshold*.

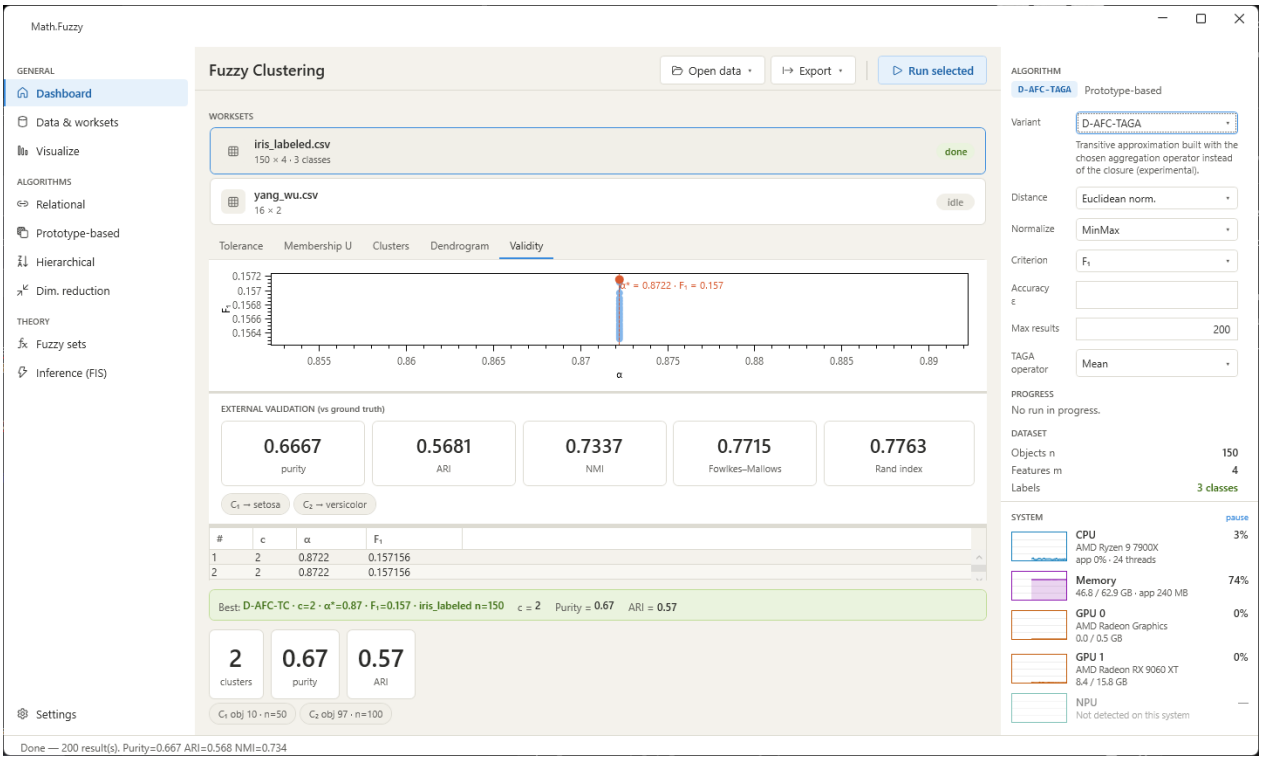


Figure 9: A TAGA variant with its field *TAGA operator*.

The category *Intuitionistic* has its own parameters; *Distance*, *Normalize*, *Criterion* and *Accuracy* are not shown for it (the criterion is F_{IFS}):

- **Clusters c** (D-AIFC(c)): the number of clusters.
- **Scan order** (all three): the order in which the pairs (α, β) are scanned, *inner* (the default) or *outer*. *outer* reproduces the published D-PAIFC result on the relation of Hung, Lee and Fuh.
- **Measure** (*Hamming* or *Euclidean*): the similarity measure that builds the tolerance from intuitionistic attribute data; it is not used when a tolerance was opened as two matrices.

The result line shows the pair (α, β) and, for D-AIFC(c), the bound w.

A parameter that is missing or out of range is reported in red under the fields, and *Run selected* does nothing until it is corrected.

> **Reproducing the experiments of the book**
>
> The original experiments on the Iris data use the squared Euclidean distance, normalisation *Max* and the accuracy $= 0.00001$. With D-AFC and c = 3 this setting gives the allotment of the monograph with six misclassified objects.

6 Running an algorithm

Select the workset, configure the panel and press *Run selected*. The workset shows *running…*, the header shows the elapsed time and *Stop*, and the progress bar reports the scan of the levels. Most runs on the sample data take well under a second; the relational algorithms on a few hundred objects can take minutes, because they enumerate combinations of centres on every level. *Stop* interrupts a run at the next level; the results of the previous run stay on the tabs.

When the run finishes, the status bar prints the number of allotments found and — for labelled data — purity, the adjusted Rand index and the normalised mutual information of the best one. If no allotment satisfies the parameters, the status bar says so; typical remedies are a different distance, *All levels* instead of a manual level, or a larger w.

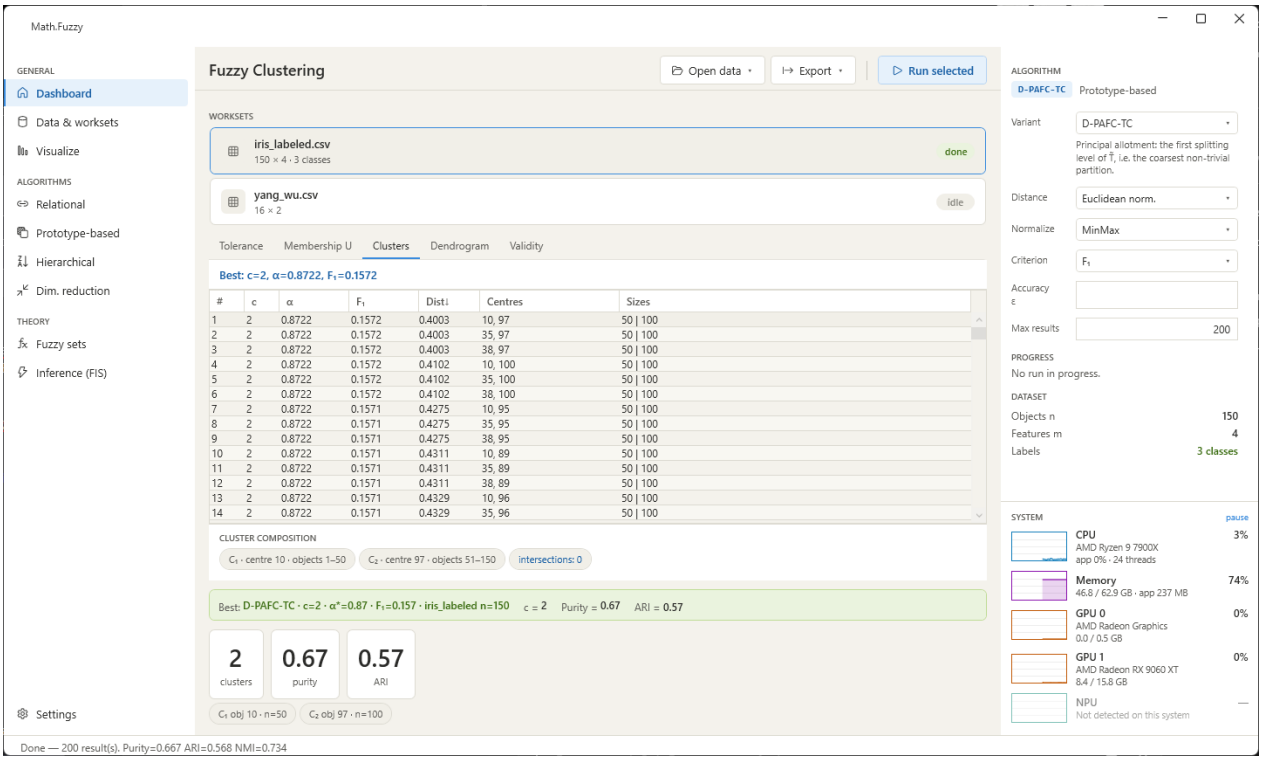


Figure 10: A D-PAFC-TC run on the Iris data has just finished.

7 Reading the results

7.1 Clusters

The table lists the allotments found, best first: the rank, the number of clusters, the level , the value of the criterion, the prototype distance used to break ties (closure-based algorithms only), the centres (typical points, 1-based object numbers) and the sizes of the clusters. Selecting a row shows its composition below the table: every cluster with its centre and the ranges of its objects, and the number of objects lying in more than one cluster.

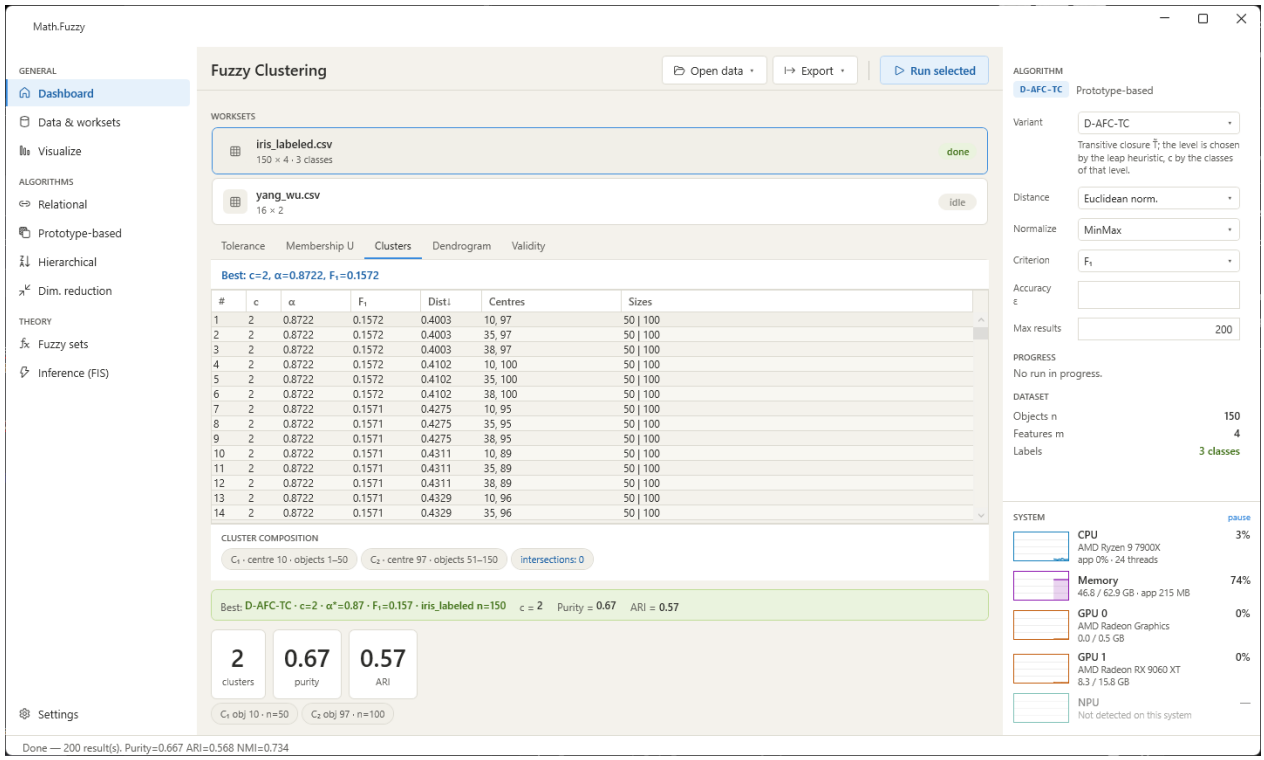


Figure 11: The list of allotments after a D-AFC-TC run.

7.2 Membership U

The membership matrix of the selected allotment as a heat map: one row per object, one column per cluster, the colour giving the membership degree μ. The *threshold* slider hides memberships below a value, *sort by cluster* orders the objects by their cluster. Clicking a row shows the memberships of that object on the right.

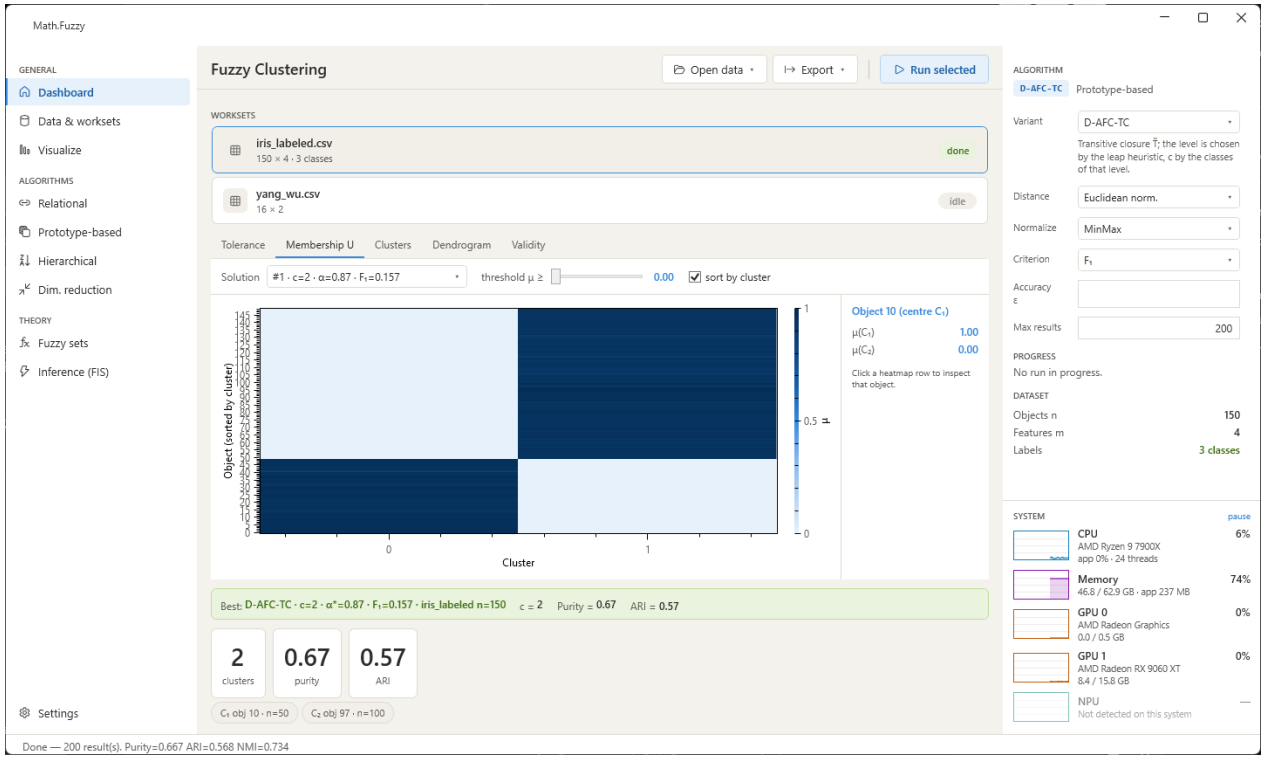


Figure 12: The membership matrix.

7.3 Tolerance

The tolerance matrix T built with the parameters of the last run, and — with the button *T (closure)* — its max-min transitive closure. The number of distinct levels is shown at the right; the *-slice* slider blanks every entry below the chosen level, which shows the crisp relation T_α at that level. Objects are in file order.

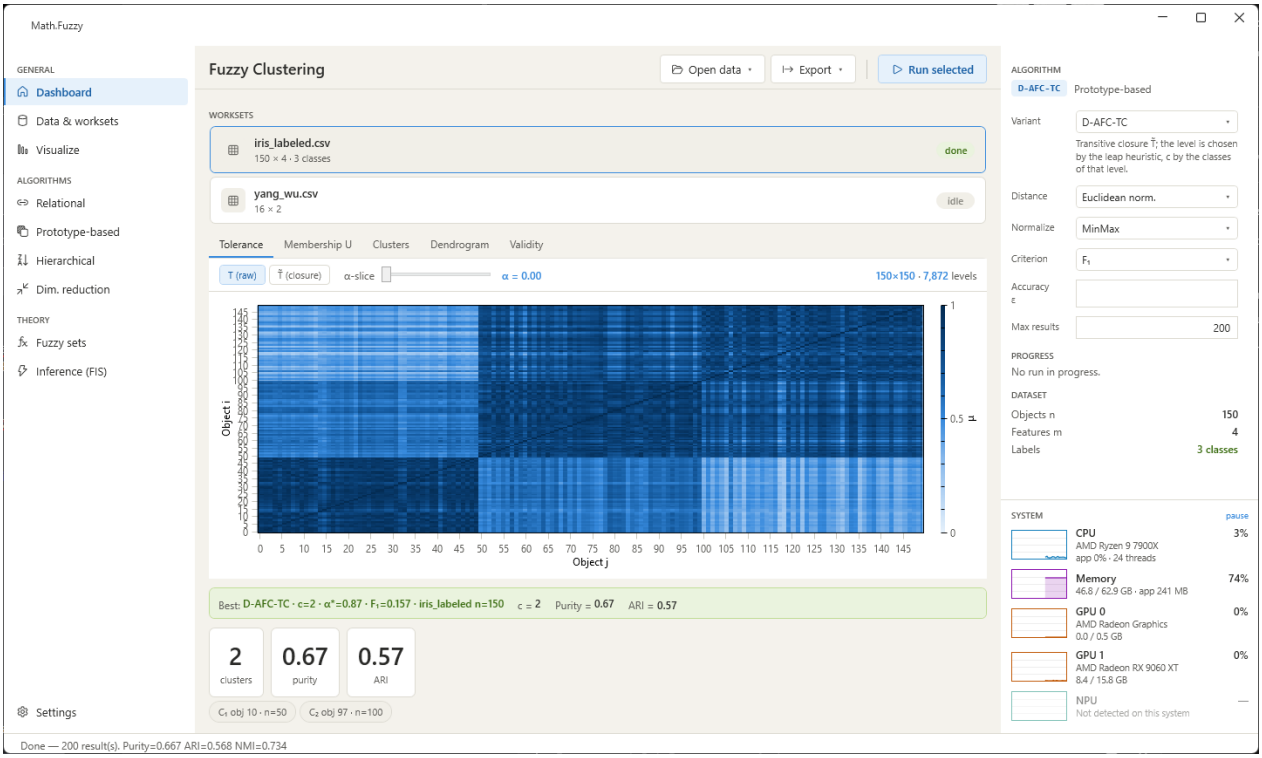


Figure 13: The tolerance matrix T of the Iris data.

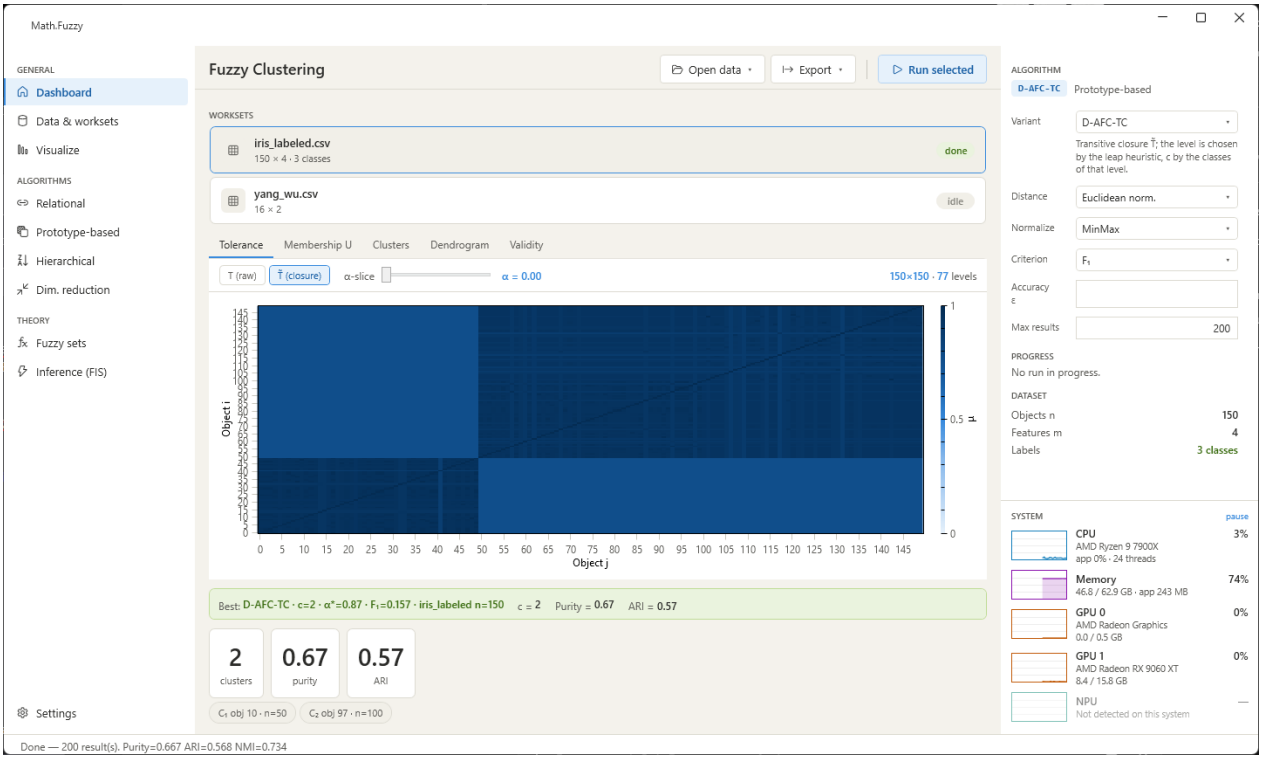


Figure 14: Its transitive closure T: the block structure of the equivalence classes.

7.4 Dendrogram

After a run of H-AFC-TC or H-AFC-TAGA the tab shows the hierarchy of allotments: the levels of the relation on the vertical axis, the finest partition at the top, merges at decreasing going down. The table on the right lists every level with its number of clusters, the cluster sizes and the criterion; clicking a level highlights its cut in the tree. For other algorithms the tab is empty.

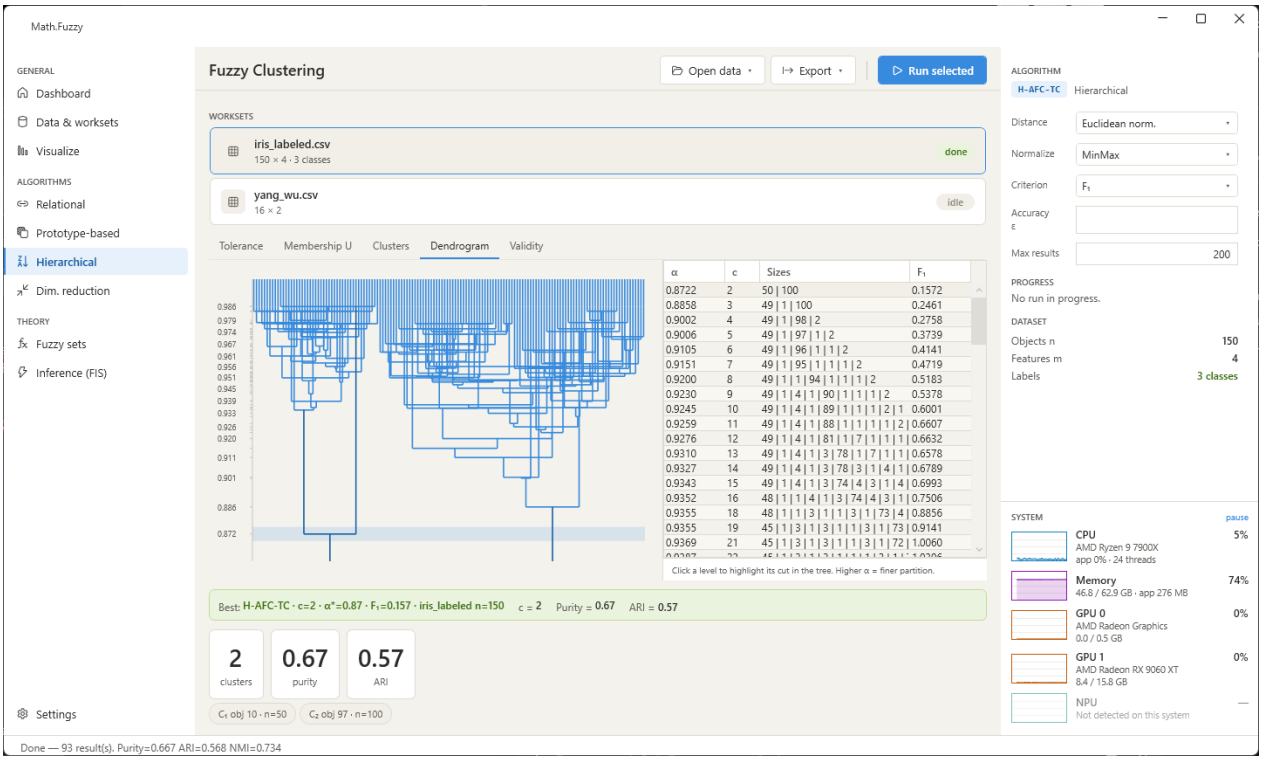


Figure 15: The dendrogram of the Iris data.

7.5 Validity

The upper plot shows the criterion of every allotment against its level, with the best one marked. For labelled data the block *External validation* gives purity, the adjusted Rand index, the normalised mutual information, the Fowlkes–Mallows index and the Rand index of the best allotment, and the majority class of every cluster.

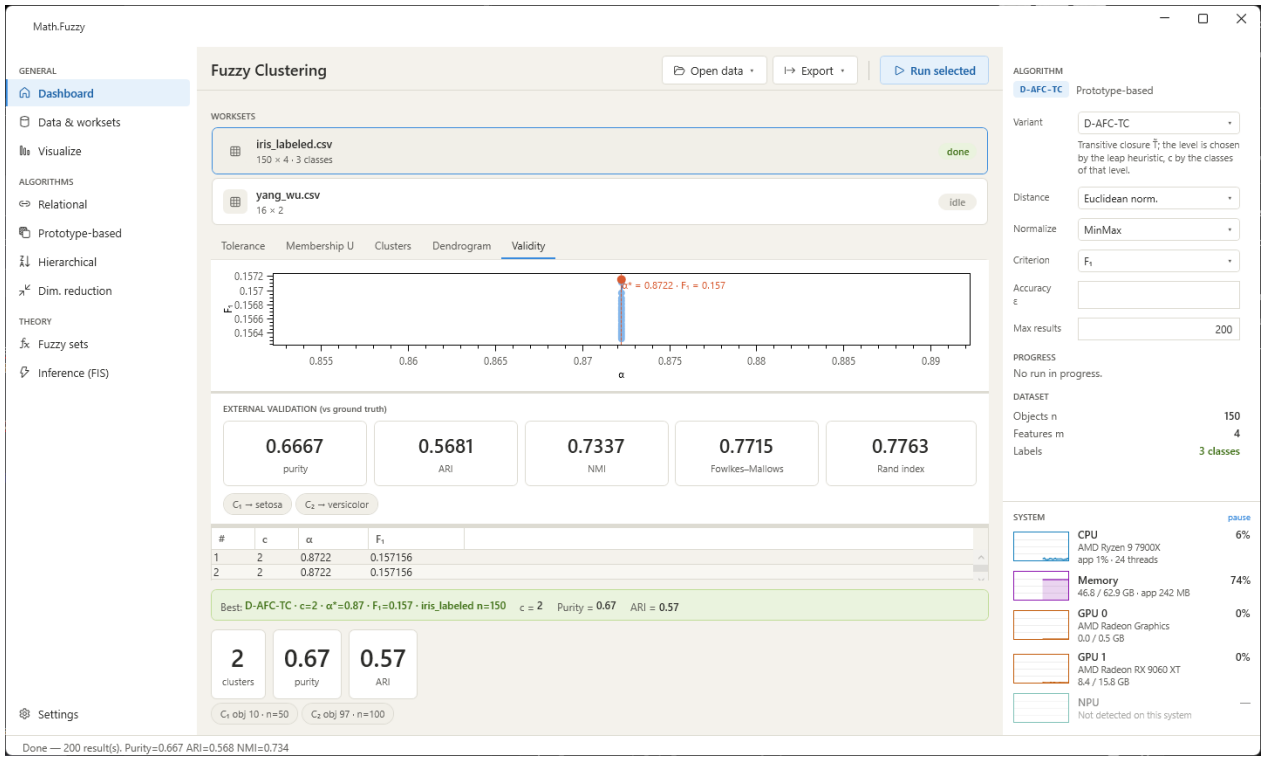


Figure 16: The validity tab after a D-AFC-TC run on the Iris data.

8 Visualize

The scatter plot shows the active workset in two chosen attributes. *Color by* switches between the clusters of the best allotment of the last run — with the typical points as black diamonds and objects of several clusters ringed — and the true classes of the file. The caption under the toolbar says which run and which data set the colours refer to.



Figure 17: The scatter plot coloured by the true classes.

9 Export

Export → Results as JSON… writes every allotment of the last run with its level, criterion, centres and clusters; object numbers are 1-based, as everywhere in the workbench. *Export → Results as CSV…* writes one row per allotment for spreadsheets. The command-line tool writes the same allotments in its own JSON layout, with 0-based object numbers and with the validity indices.

10 Theory pages

10.1 Fuzzy sets

A small laboratory for membership functions. *Shapes* draws one fuzzy set of a chosen shape (triangular, trapezoidal or Gaussian); the parameters can be typed or the square handles dragged on the plot, and the panel reports the height, support, core, -cut, crossover points, cardinality and centroid. *Operations* combines two sets with a t-norm, an s-norm or a complement, with every operator of the chosen family, and defuzzifies the result by five methods. *From cluster* shows how a fuzzy cluster of the last run becomes a term of the antecedent of a rule.

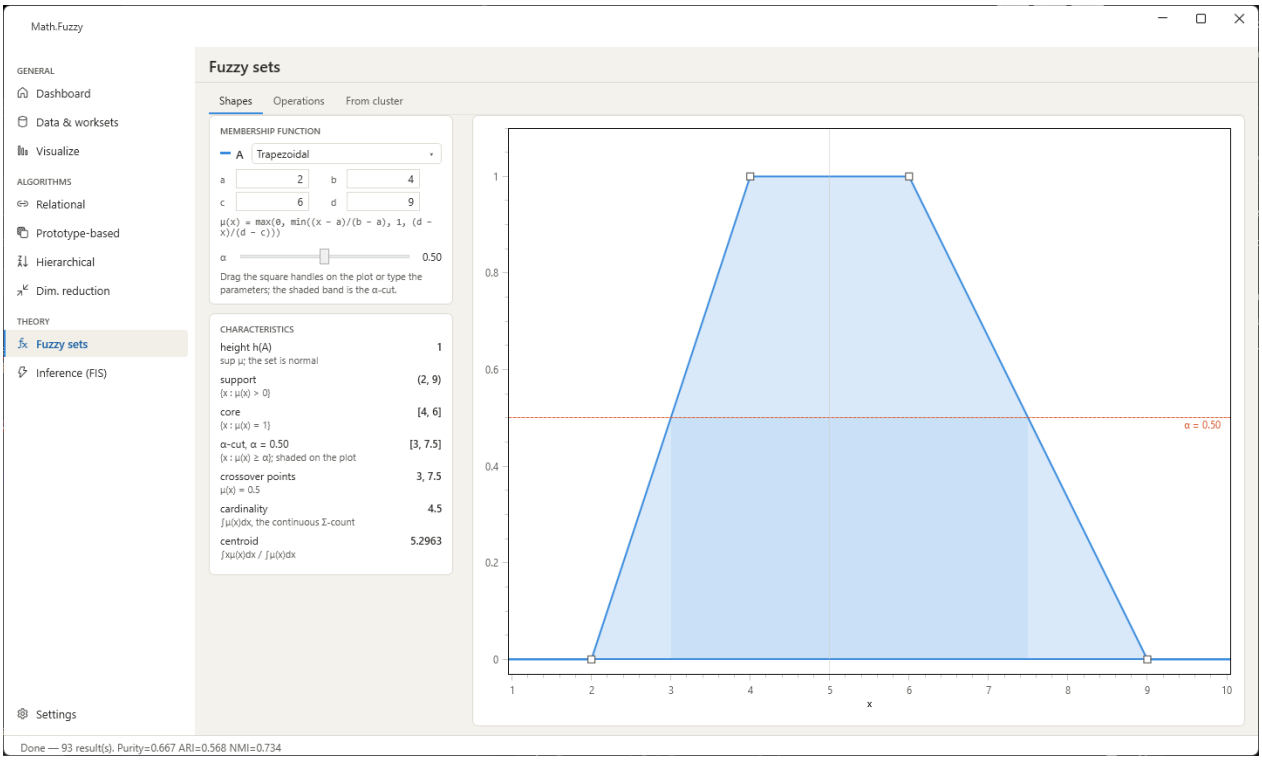


Figure 18: The Shapes tab.

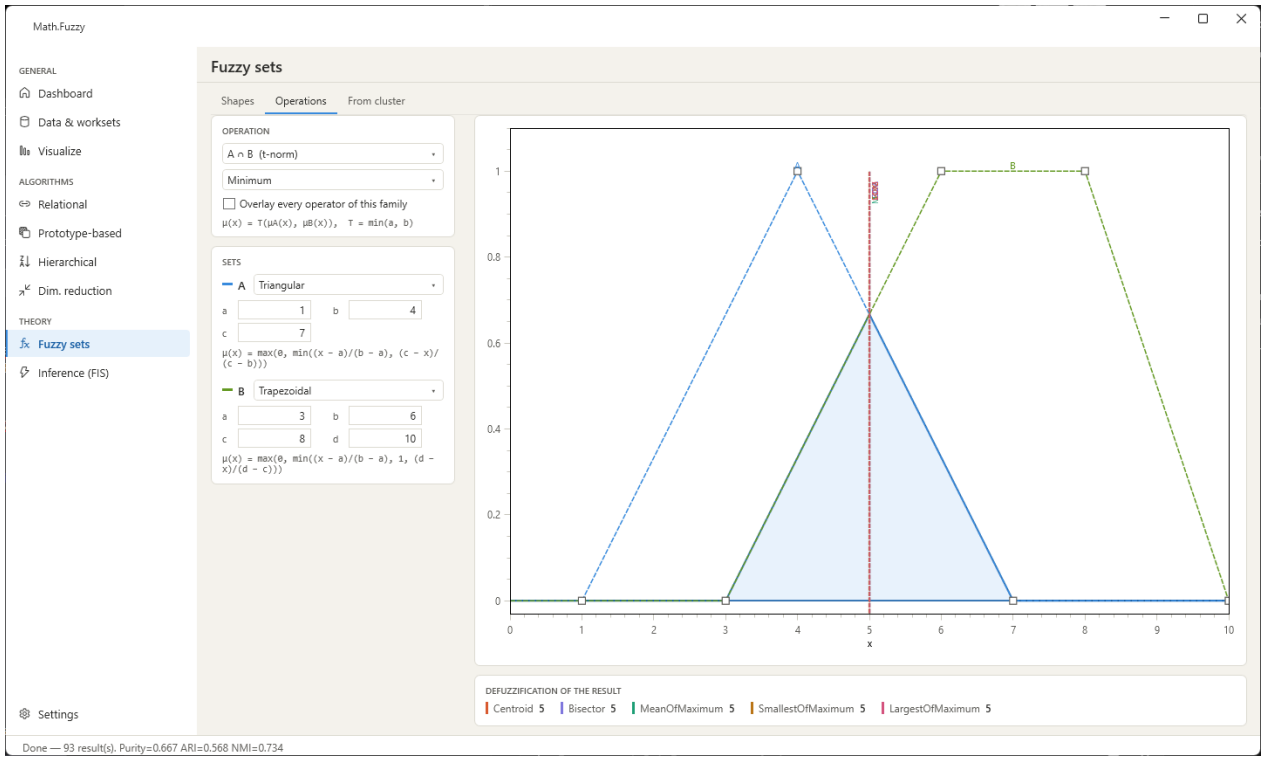


Figure 19: The Operations tab: the intersection of two sets.

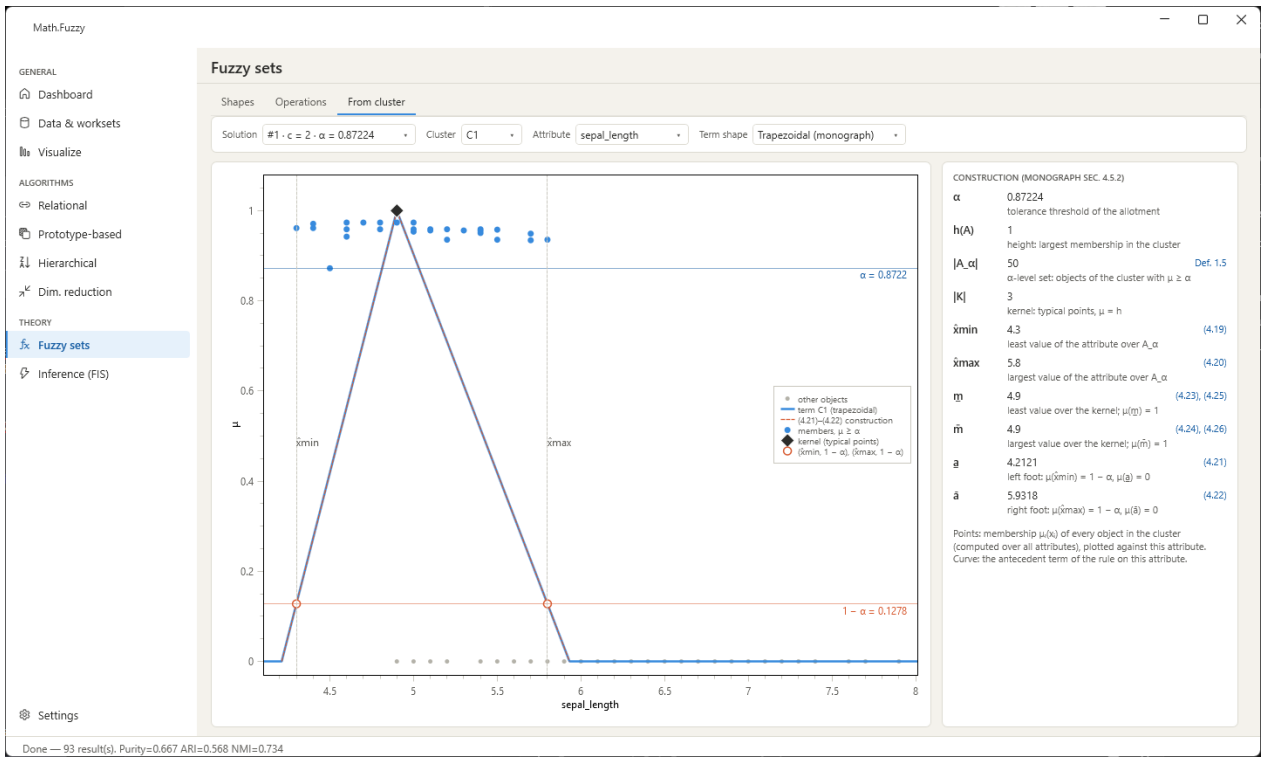


Figure 20: The From cluster tab: the antecedent term built from cluster C1 of a D-AFC-TC solution, with the construction values on the right.

10.2 Inference (FIS)

The inference page generates a fuzzy rule base from a clustering solution and lets you experiment with it.

- **Solution** picks an allotment of the last run; **Rule base** the kind of system: a Mamdani system with one output per cluster (the construction of the book), a classifier whose consequents are the class labels,

a Mamdani regression system, or a Takagi–Sugeno system of order 0 or 1 (the last three predict the column chosen in **Target**); **MF shape** the shape of the terms. *Regenerate* builds the rules.

- **Rules.** One rule per cluster. Each rule can be switched off, its terms and weight changed; the bar at the right shows its firing strength for the current input.
- **Membership functions.** The terms of one variable; parameters can be typed or dragged.
- **Input vector.** Sliders for every attribute; *Load* copies an object of the data set into them.
- **Output.** The decision, the output values and the aggregated output set with its defuzzified value.
- **Evaluation on the workset.** The accuracy of the rule base on all objects and its confusion matrix.

The rule base can be exported as JSON or as a MATLAB Fuzzy Logic Toolbox file (`.fis`); a JSON file can be imported back, and evaluated without the workbench by `mfuzzy infer`.

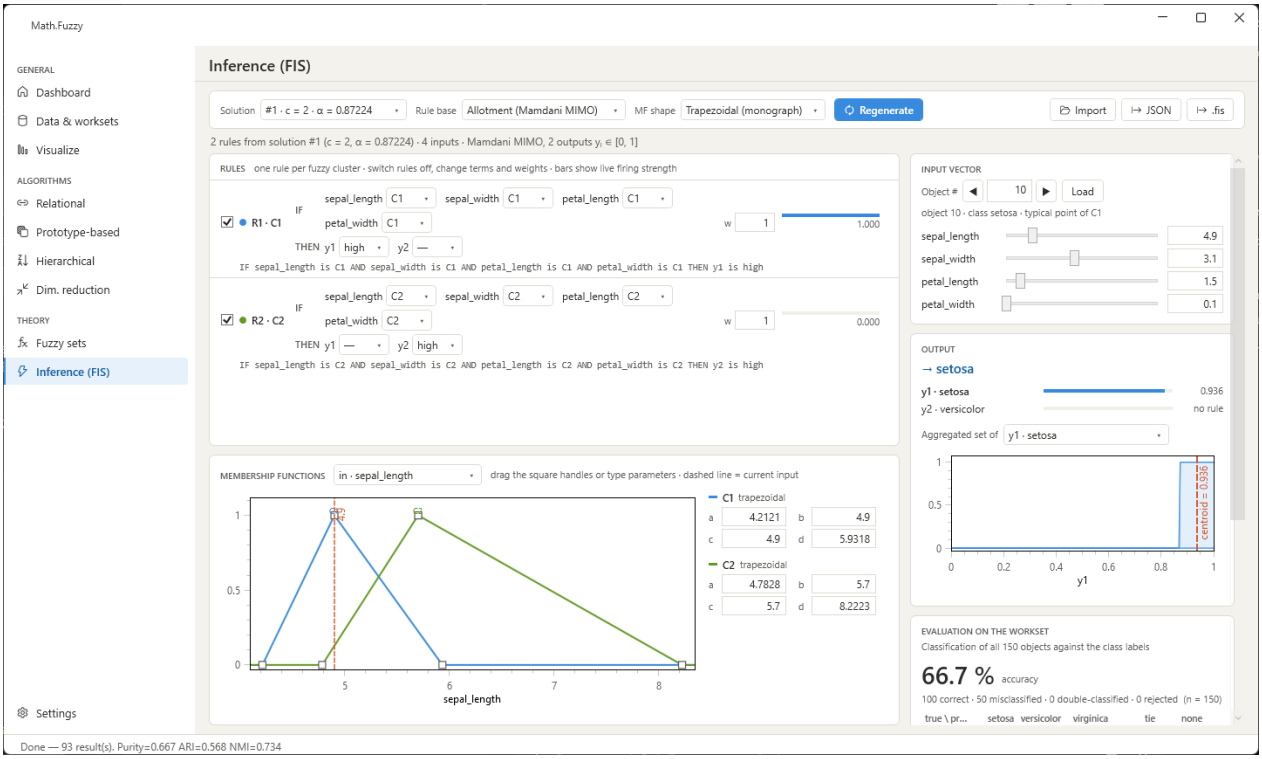


Figure 21: The inference page with a rule base generated from a D-AFC-TC solution.

11 Settings

The page sets the defaults for the number of results kept and the minimal size of a cluster (applied by every algorithm), and shows the versions of the components.

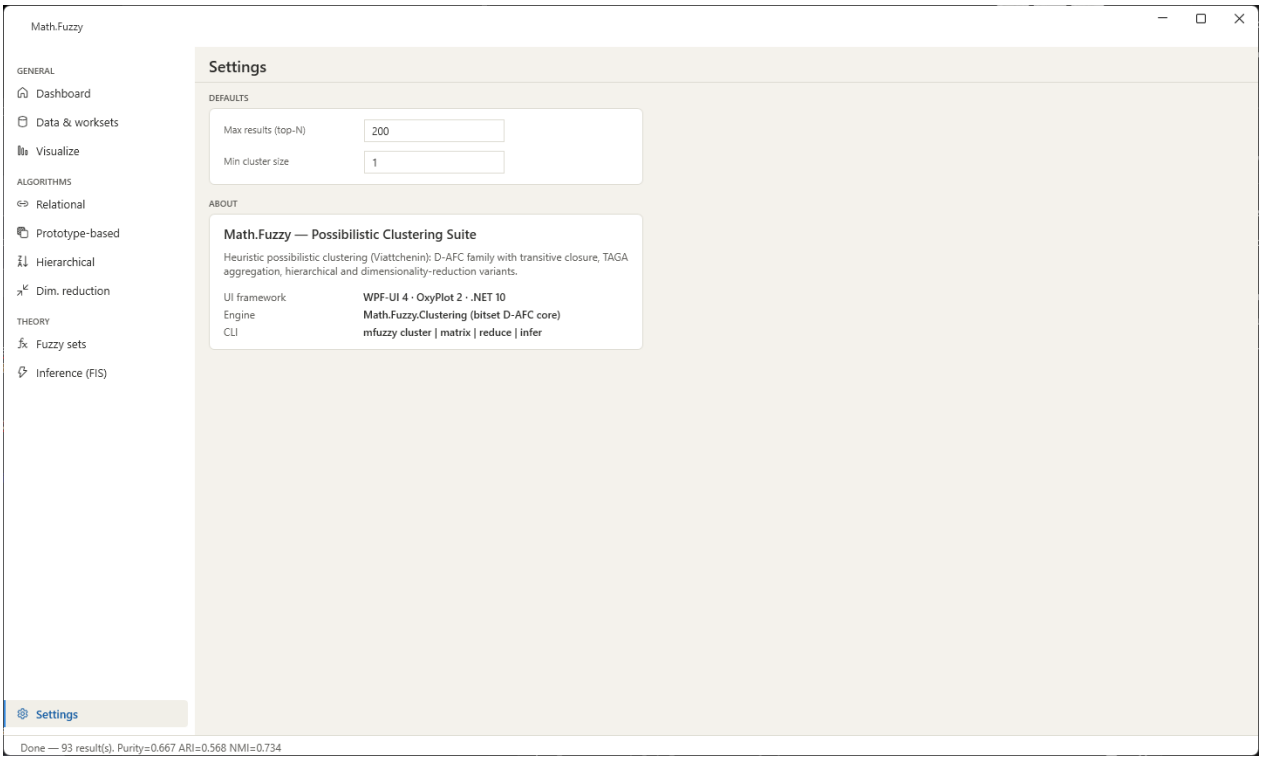


Figure 22: The Settings page.

12 Troubleshooting

- *Windows refuses to start the application.* SmartScreen: choose *More info* → *Run anyway*. The checksum of the archive can be verified against the `.sha256` file published with the release.
- *"D-AFC needs the number of clusters c."* Enter an integer of at least 2 in *Clusters c*, or choose D-PAFC to have the number searched.
- *"No valid clustering found."* No allotment satisfies the parameters. With a manual level try *All levels*; with D-PAFC increase w; try another distance.
- *A run takes minutes.* The relational algorithms enumerate centre combinations on every level; a few hundred objects with $c = 3$ can take several minutes, and partially separated clusters ($w > 0$) rescan all levels. The closure-based algorithms are fast. *Stop* interrupts a run.
- *"The search stopped at its node budget."* A relational run visited more combinations than its budget allows. The allotments shown are the best found so far and better ones may exist: restrict the search (a manual level, an accuracy threshold) or use the command-line tool with a larger `--max-nodes`.
- *A dimensionality-reduction run on a matrix workset is refused.* Attribute selection needs attributes; open the data file instead of the matrix.
- *The results of a previous run are still shown after an error.* They belong to the previous run; the status bar reports the error of the new one.

13 The command-line tool

The clustering runs of the workbench can be reproduced from a terminal with the tool `mfuzzy` of the same repository, for example

```
mfuzzy cluster --algorithm dafc --input datasets/iris_labeled.csv --clusters 3 ^
    --distance squared-euclidean --normalize max --accuracy 0.00001
```

The options carry the names of the panel fields (`--distance`, `--normalize`, `--criterion`, `--accuracy`, `--clusters`, `--alpha`, `--max-size`, `--intersections`, `--taga`; `--matrix` for a tolerance matrix), and `--format json` writes the same JSON as the *Export* command. The README of the repository documents the tool, the .NET library and the MATLAB toolbox. The intuitionistic algorithms `dpaifc`, `daifc` and `dpaifc-tc` take either two matrices (memberships with `--input`, non-memberships with `--nonmembership`) or one file

of intuitionistic attribute data in pairs of columns (`--ifs-measure hamming` or `euclidean`); `--scan-order beta-inner` or `beta-outer` sets the order of the scan. For example

```
mfuzzy cluster --algorithm daifc --clusters 3 --input datasets/hung_ifs_membership.csv ^
    --nonmembership datasets/hung_ifs_nonmembership.csv
mfuzzy cluster --algorithm dpaifc-tc --input datasets/cars10_ifs.csv
```

find three clusters at $(\alpha, \beta) = (0.4, 0.6)$ with $w = 1$ and four clusters at $(0.4833, 0.1667)$. The table has an extra column *Beta*, and the options `--distance`, `--normalize`, `--criterion` and `--accuracy` are not used.

The tool also has the commands `size` (choice of the largest cluster size u), `bounds` (bounds of the number of clusters), `matrix build`, `matrix closure`, `reduce` and `infer`; the README documents them.